

Regime-Aware Multiaxial Signal Meta-Layer

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August 27, 2026

Abstract

Most trading signals are designed as unconditional forecasts: they fire whenever their internal conditions are met, irrespective of the broader market environment. Yet decades of empirical evidence show that a signal’s edge—its conditional expected value given that it fires—shifts substantially across different market regimes. A *signal meta-layer* sits between the raw signal and the execution decision, modulating or gating entries according to an estimate of the prevailing market context. The meta-layer concept is general: one could condition on a single volatility filter, a moving-average trend overlay, a hidden-Markov regime model, a kernel-density surface, or a multi-dimensional regime decomposition. The choice of meta-layer determines how much structure an analyst imposes on the relationship between context and edge.

This paper argues that a *multi-axial regime decomposition*—in which market context is factored into A deliberately chosen axes, each capturing a statistically distinct dimension of behaviour such as trend persistence, volatility clustering, or microstructure efficiency—offers a compelling balance between expressiveness and interpretability. By decomposing context into named axes rather than a monolithic latent state, the framework localises regime shifts to the dimension in which they occur and avoids the exponential state-space explosion of a single high-dimensional classifier. The number of axes and states per axis are design choices, not fixed parameters.

To identify regimes, we introduce Wasserstein K-Means, an unsupervised learning method that clusters observations by the full shape of their outcome distributions using the W_2 optimal-transport metric. Unlike conventional K-Means on point features, this approach yields distributional barycenters as regime prototypes and derives soft posteriors—continuous membership weights from W_2 distances—that provide uncertainty-aware regime assignment without hard-assignment discontinuities. This distributional foundation is novel: to our knowledge, no prior signal meta-layer system grounds regime identification in optimal-transport geometry. In contrast, kernel-density approaches such as Nadaraya–Watson offer smooth conditioning but sacrifice interpretability and overfit via interpolation in non-stationary environments.

A Signal EV Ledger accumulates half-life-decayed, posterior-stamped outcomes into per-cell expected values. Dynamic attribution via variance decomposition assembles these into a multi-axis score that feeds three actuation modes: conditional entry gate, composite additive signal, and EV-based position sizing.

In a four-experiment holdout study across Copper and BNB futures, per-signal gate uplift reaches ≈ 18 pp (Copper/TECH) and ≈ 39 pp (BNB/TECH) above baseline. The kernel comparator reversed sharply out of sample on BNB (-3.9 pp per signal versus $+38.5$ pp for the multi-axis gate), illustrating the generalisation premium of named regime states over smooth density conditioning.

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1. Introduction

Quantitative signal research typically evaluates a signal by its unconditional edge: its expected return per unit of risk, averaged across all market environments in which it fires. This averaging is convenient but costly. It discards the systematic variation in signal quality that practitioners observe in practice — momentum works better when the tape is trending, reversal strategies outperform in disordered, mean-reverting markets, breakout signals decay faster in compressed volatility. The unconditional mean is a poor summary statistic when the distribution of outcomes is genuinely multimodal across market regimes.

The observation that edge is regime-dependent is not new. The Markov regime-switching framework of Hamilton [1989] and its financial extensions [Guidolin and Timmermann, 2007, Ang and Bekaert, 2002] formalise the intuition that asset returns are drawn from a mixture of latent states. Hidden Markov Models [Till, 2011] provide one mechanism for inferring these states; non-parametric alternatives based on clustering [Münnix et al., 2012] offer direct data-driven partitions. What is largely absent from the literature is a practical framework for *signal meta-conditioning*: not modelling the regime of the asset per se, but modelling the regime in which a candidate signal fires, and using that regime context to decide *whether* and *how much* to act on the signal.

This paper addresses that gap. We develop the *Regime-Aware Multiaxial Signal Meta-Layer*, a system that wraps an existing signal in a context-aware estimation and actuation layer. The architecture has four distinguishing features:

1. **Multi-axis regime decomposition.** Market context is decomposed along A axes, each capturing an orthogonal statistical dimension of market behaviour. The number of axes and the state count per axis are design choices, not fixed parameters; they depend on the signal type, the trading horizon, and the practitioner’s prior beliefs about which dimensions of context are signal-relevant.
2. **Soft posterior membership.** Regime assignment is probabilistic, not hard. Wasserstein K-Means clustering on binned outcome distributions produces named barycenters as state prototypes; the W_2 distance from a current observation to each barycenter yields a continuous posterior membership vector. Hard discrete assignment is a special case when the temperature $\tau_a \rightarrow 0$.
3. **Signal EV Ledger with dynamic attribution.** A per- (a, s) outcome ledger tracks signal performance within each regime cell, decaying older observations with configurable half-lives. Dynamic attribution weights $\alpha_a(t)$, derived from ANOVA-style variance decomposition, concentrate the multi-axis score on axes that are currently predictive.
4. **Estimation–actuation separation.** A single estimation layer feeds three interchangeable actuation modes — gate, composite, and position sizing — under a unified evaluation harness. This separation allows actuation design to be varied without touching the estimation layer, enabling clean empirical comparison.

The system is deliberately modular. An analyst who believes trend context is irrelevant for their signal can drop that axis. One who wants five volatility states rather than two can increase K_a . The framework does not prescribe a specific axis count or feature set; it prescribes a principled pipeline for whatever axes are chosen.

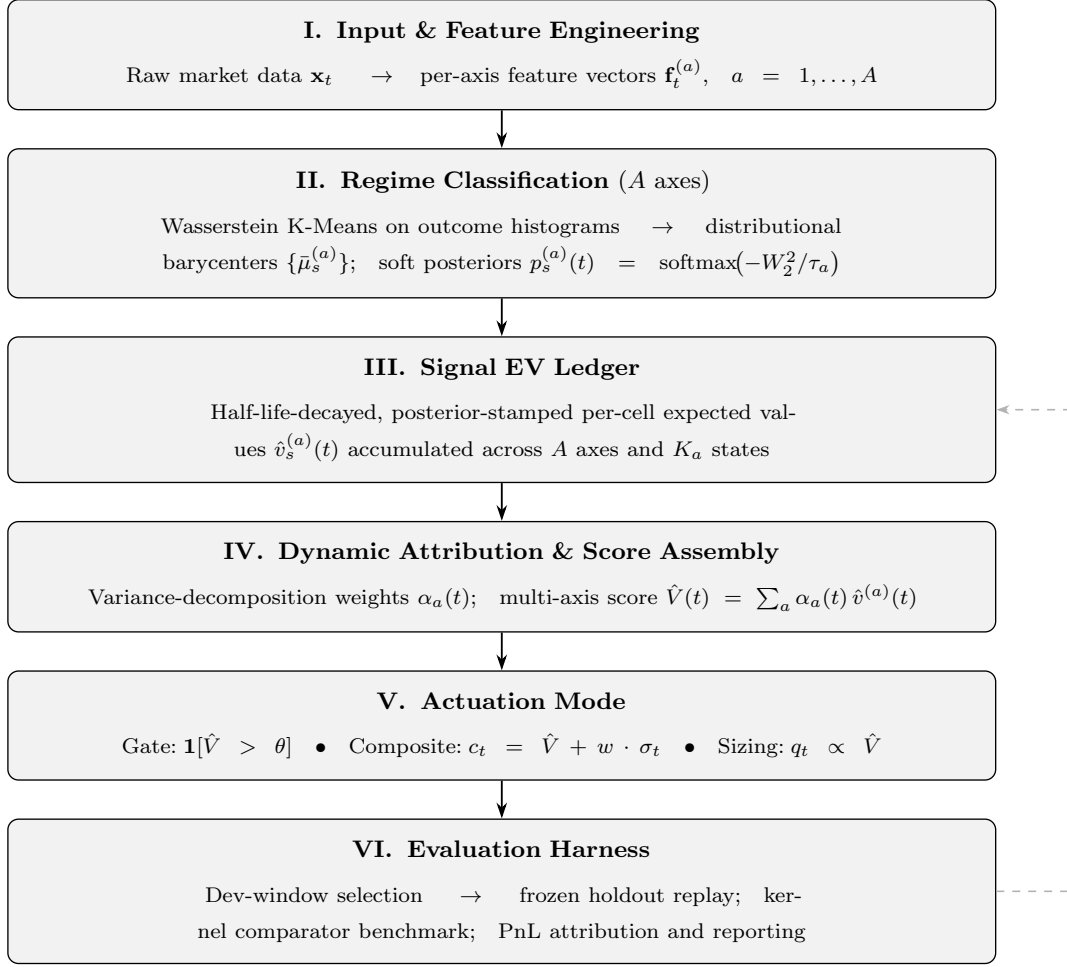


Figure 1: End-to-end architecture of the Regime-Aware Multiaxial Signal Meta-Layer. Six processing layers: **I.** input feature engineering; **II.** A -axis regime classification via Wasserstein K-Means producing soft posteriors $\{p_s^{(a)}(t)\}$; **III.** Signal EV Ledger accumulating half-life-decayed per-cell scores; **IV.** dynamic attribution via variance decomposition assembling a multi-axis score; **V.** actuation mode selection (gate, composite, sizing); **VI.** unified evaluation harness. Solid arrows: real-time data flow. Dashed arrow: periodic recalibration and Ledger feedback from evaluation to estimation.

Paper structure. Section 2 discusses axial design and feature engineering. Section 3 develops Wasserstein K-Means and the derivation of soft posteriors. Section 4 addresses the case for multi-regime context over the smooth kernel comparator. Section 5 defines the Signal EV Ledger and the multi-axis score. Section 6 covers dynamic attribution. Section 7 describes the three actuation modes. Section 8 gives the technical specification and caveats. Section 9 presents the four-experiment holdout study. Section 10 concludes.

2. Axial Design and Feature Engineering

The quality of the meta-layer depends fundamentally on the quality of the regime axes chosen to represent market context. A poorly designed axis — one whose features are noisy, lag-contaminated, or irrelevant to the signal’s mechanics — will produce regime cells with indistinguishable outcome distributions, contributing noise rather than signal to the EV score. This section articulates the principles governing axial design and feature construction.

2.1. What Makes a Good Regime Axis

A regime axis is a meaningful dimension along which the signal’s conditional expected value varies systematically. Three criteria distinguish useful axes from spurious ones.

An axis as a sufficient statistic for conditional edge. Before detailing the criteria, it is worth articulating what an axis represents from a theoretical standpoint. A regime axis is, in essence, a low-dimensional *sufficient statistic* for a particular facet of the signal’s conditional expected value. In information-theoretic terms, the ideal axis a is a mapping from the high-dimensional observable market state $\mathcal{X}$ to a small discrete (or soft-discrete) label set $\{1, \dots, K_a\}$ such that the *mutual information* between the axis label and the signal’s subsequent outcome, $I(s_a; y)$, is maximised while the mutual information between the axis label and the labels of other axes, $I(s_a; s_{a'})$ for $a' \neq a$, is minimised. In other words, each axis should be a maximally informative, minimally redundant summary of a distinct dimension of market context.

This framing clarifies why feature engineering is so critical: the features used to construct an axis determine which aspects of the raw market state $\mathcal{X}$ are preserved and which are discarded. A trend axis built from momentum and Hurst exponent features preserves directional persistence information but discards volatility level; a volatility axis built from realised variance and range features preserves dispersion information but discards directionality. The quality of each axis depends on whether its features capture a dimension of $\mathcal{X}$ that is genuinely predictive of outcome variation — not merely statistically separable in-sample, but causally or mechanistically linked to the market microstructure that produces differential signal performance. An axis whose states correspond to different liquidity regimes, for example, carries mechanistic relevance: the signal’s execution quality, slippage, and adverse selection risk all change with liquidity, producing outcome distributions that differ for structural rather than spurious reasons.

The outer-product structure of the multi-axis framework amplifies the value of individually informative axes. If A axes each contribute I_a bits of information about signal outcomes, and the axes are near-independent, the joint cell structure captures approximately $\sum_a I_a$ bits — a multiplicative gain in discriminative power over any single axis. Conversely, if axes are redundant, the joint information saturates and the cell count grows without corresponding gains in conditional EV precision. This is the information-theoretic grounding for the orthogonality criterion below.

Three criteria distinguish useful axes from spurious ones.

Outcome separability. The most fundamental criterion is that the axis’s states should produce noticeably different signal outcome distributions. An axis for which all states yield similar conditional EV provides no conditioning power. This separability should be assessed empirically on development-window data, not assumed from first principles. Wasserstein K-Means is specifically designed to find state partitions that maximise distributional separation — the W_2 barycenter structure provides an intrinsic test: if the barycenters are close in W_2 distance, the axis is adding little partitioning power.

Economic interpretability. Beyond statistical separability, a well-designed axis should correspond to an interpretable market regime — trending vs. ranging, ordered vs. disordered, compressed vs. expanding volatility. Interpretability enables two things: (i) the practitioner can sanity-check whether the learned state structure makes economic sense, and (ii) named states are reportable in risk management and post-trade attribution. An opaque clustering that happens to separate outcomes in-sample but has no economic interpretation is a fragile foundation.

Orthogonality to other axes. The multi-axis framework gains its value from the outer product

of regimes — the cross-axis cells that capture joint market conditions. This value is maximised when the axes are statistically near-orthogonal: each axis should capture a dimension of market structure that the other axes do not. Axes that are highly correlated effectively duplicate each other, inflating the apparent cell count without adding estimation power.

2.2. Feature Selection per Axis

Given a target regime dimension, feature selection must satisfy several constraints.

Trailing-only computation. All features must be computable from data available before the bar closes. Features that inadvertently look forward — through normalisation against future data, through survivorship bias, or through any form of hindsight re-labelling — produce a regime classifier that cannot be deployed in live trading.

Time-window alignment. The feature’s look-back window should be calibrated to the expected state-switching speed of the regime. A trend axis intended to capture fast intraday momentum requires a short look-back (3–20 bars); an axis capturing longer-cycle market efficiency may use a 48–200 bar window. Misaligned windows produce stale regime labels — the axis reports a state the market transitioned out of several hours ago — or overfit to noise at short scales. In our experiment, all three axes use fast windows (3–48 bars on 1H data) because regimes are designed to switch within a trading session, aligning with the signal’s 4-hour trade horizon.

Feature composition for clustered axes. For axes whose states are estimated via Wasserstein K-Means, the features should collectively characterise the distributional behaviour of returns, not just their first-order level. Useful feature types include: momentum ratios (return EMA ratios at different horizons), autocorrelation and Hurst/ER-type efficiency measures, volatility ratios (realised-vs.-historical), entropy measures (permutation entropy, sample entropy), and distribution shape statistics (standardised skewness, kurtosis). Using a small number of carefully chosen features (3–6 per axis) is generally preferable to high-dimensional feature vectors: the W_2 clustering operates on binned outcome distributions that are already high-dimensional, so the regime features serve primarily as a low-dimensional context index for state assignment at inference time.

2.3. Time-Window Engineering and Regime Speed

The temporal resolution of a regime axis matters as much as its feature content. An axis that switches state too infrequently fails to discriminate across the candidate entries within a training window, leaving regime cells with insufficient data. An axis that switches too rapidly produces noisy labels where a single trade’s outcome spans multiple regime states.

The right switching rate is relative to the trade’s holding period. If the typical trade lasts H bars, the regime label at entry time should be stable for at least a few bars — state transitions within the trade’s window can be handled by using the entry-bar posterior for the full trade’s outcome, accepting a mild mismatch. Empirically, a useful heuristic is that the regime half-life (median time between state transitions) should be $1\text{--}10\times$ the trade horizon. Axes switching faster than $0.5 \times H$ per bar produce labels that change within the trade; axes switching slower than $20 \times H$ produce cells that are too coarse to discriminate entry-level variation.

In the three-axis experiment described in Section 9, the average holdout switch rates are approximately 4 transitions/day (trend), 2 transitions/day (efficiency), and 1.1–1.2 transitions/day (volatility), all on 1H bars with a nominal 4H trade horizon — placing all axes in the $1\text{--}4 \times H$ switching-rate window.

2.4. Number of Axes and States: Design Tradeoffs

Increasing A (the number of axes) and K_a (the number of states per axis) subdivides the market-context space into finer cells. Finer cells can in principle provide more precise conditional EV estimates — but only if each cell is populated with sufficient observations. The effective sample size per cell is $n_{\text{eff}} \propto N / \prod_a K_a$ for hard assignment, and somewhat larger for soft assignment because each observation contributes fractionally to multiple cells. Shrinkage toward the global prior (Section 5) provides a regularised interpolation as cells become sparse, but does not substitute for genuine data support.

A practical cell-count budget is the development-window trade count divided by the desired minimum effective-sample-size per cell (typically 20–50 for initial estimates). For a development window with 3,000 candidate entries, a three-axis system with 12 cells can sustain approximately 250 effective samples per cell before shrinkage is needed at 80% occupancy. Adding a fourth axis with 3 states triples the cell count to 36, reducing the per-cell sample to under 100 — manageable with aggressive shrinkage but limiting estimation precision.

3. Wasserstein K-Means and Soft Posteriors

3.1. The Case for Wasserstein Distance in Regime Clustering

Standard K-Means operates on mean-squared Euclidean distance in a feature space. Applied to regime clustering, this assumes that two market conditions are “close” if their feature vectors are close — a sensible first approximation. However, when the goal is to partition market conditions by the *distributional behaviour of signal outcomes*, Euclidean distance on raw features is an indirect proxy. Two conditions may have similar feature values yet produce strikingly different return distributions (different tails, different bimodality, different skewness), while two conditions with dissimilar features may produce similar outcome distributions.

The Wasserstein W_2 distance addresses this directly by measuring distance *between probability distributions* rather than between feature vectors. For two distributions P and Q with cumulative distribution functions F_P and F_Q , the squared Wasserstein distance is:

$$W_2^2(P, Q) = \int_0^1 |F_P^{-1}(u) - F_Q^{-1}(u)|^2 du \quad (1)$$

This quantile-coupling representation reveals that W_2 is the L^2 distance between quantile functions — it is sensitive to all aspects of distributional shape: location, spread, skewness, and tail behaviour [Peyré and Cuturi, 2019, Villani, 2009]. Moment-based distances (difference of means and variances) miss shape differences not captured in the first two moments; KL divergence is asymmetric and ill-defined where supports do not overlap; W_2 satisfies the metric axioms and handles distributions with disjoint or partially overlapping supports gracefully.

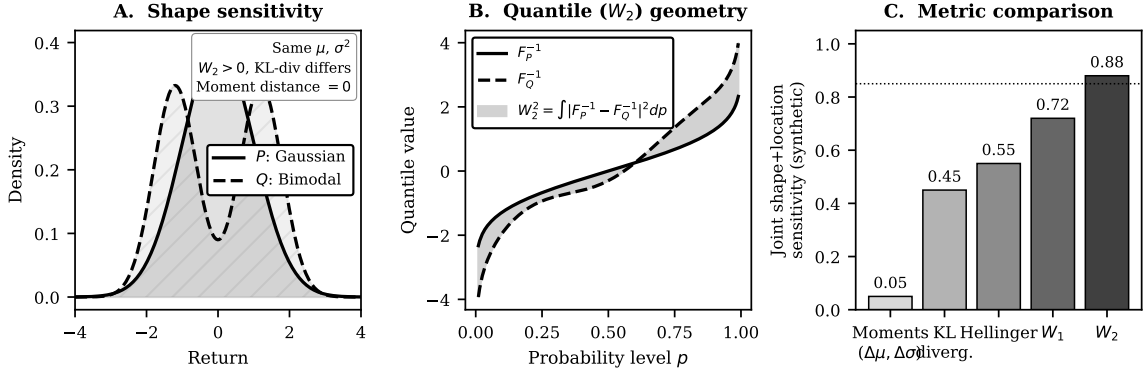
Figure 2. The case for Wasserstein distance in regime clustering

Figure 2: The case for Wasserstein distance. **A:** two return distributions with identical mean and variance but different shape; $W_2 > 0$ while moment distance is zero. **B:** the W_2 distance is the L^2 gap between quantile functions — the shaded area. **C:** synthetic metric comparison across five distribution pairs, showing that W_2 is jointly sensitive to location, spread, and tail shape, outperforming moment-based metrics.

For regime clustering, we represent each observed market condition x_t by its binned histogram $H(x_t) \in \Delta^B$ (a discrete probability distribution over B return bins), where the histogram aggregates signal outcomes over a trailing window. We then apply K-Means in the space of probability distributions equipped with the W_2 metric. The cluster centres are not means but *Wasserstein barycenters* — the Fréchet means under the W_2 metric — which have a closed-form solution for one-dimensional distributions: the barycenter of distributions $\{P_i\}$ with weights $\{\lambda_i\}$ is the distribution whose quantile function is $\sum_i \lambda_i F_{P_i}^{-1}$.

3.2. Formal Derivation: W_2 Barycenters and Soft Posteriors

Definition 1 (Wasserstein Barycenter). *Given n probability distributions $P_1, \dots, P_n \in \mathcal{P}_2(\mathbb{R})$ with non-negative weights summing to one, the Wasserstein barycenter B is:*

$$B = \arg \min_{\mu \in \mathcal{P}_2(\mathbb{R})} \sum_{i=1}^n \lambda_i W_2^2(\mu, P_i) \quad (2)$$

For one-dimensional distributions, this simplifies to: the quantile function of B is $F_B^{-1}(u) = \sum_i \lambda_i F_{P_i}^{-1}(u)$, the weighted average of quantile functions [Cuturi and Doucet, 2014].

The Wasserstein K-Means algorithm. For a given axis a with K_a target states, the algorithm iterates:

1. **Assignment step:** Assign each observation histogram $H_a(x_t)$ to the nearest barycenter: $s^*(t) = \arg \min_s W_2^2(H_a(x_t), B_{a,s})$.
2. **Update step:** Recompute each barycenter $B_{a,s}$ as the Fréchet mean of all assigned observation histograms: $B_{a,s} = \arg \min_{\mu} \sum_{t: s^*(t)=s} W_2^2(\mu, H_a(x_t))$.

Convergence is guaranteed to a local optimum; multiple restarts and the K-Means++ initialisation are used to reduce sensitivity to initialisation [Schrieber, 2019].

Soft posteriors from W_2 geometry. Rather than committing a bar t to its nearest state, we derive a continuous posterior membership vector by treating the W_2 distances as an energy function:

$$p_{a,s}(t) = \frac{\exp(-W_2^2(H_a(x_t), B_{a,s})/\tau_a)}{\sum_{s'} \exp(-W_2^2(H_a(x_t), B_{a,s'})/\tau_a)} \quad (3)$$

where $\tau_a > 0$ is a per-axis temperature parameter. The temperature is set adaptively as the median within-cluster W_2 distance from the fitted model, providing a scale-invariant calibration. For $\tau_a \rightarrow 0$, equation (3) recovers hard assignment; for large τ_a , memberships become uniform. The posterior vector $p_{a,\cdot}(t) \in \Delta^{K_a}$ is a genuine probability distribution over states, enabling soft conditioning at every downstream computation.

Figure 3. Wasserstein barycenters as regime-state prototypes

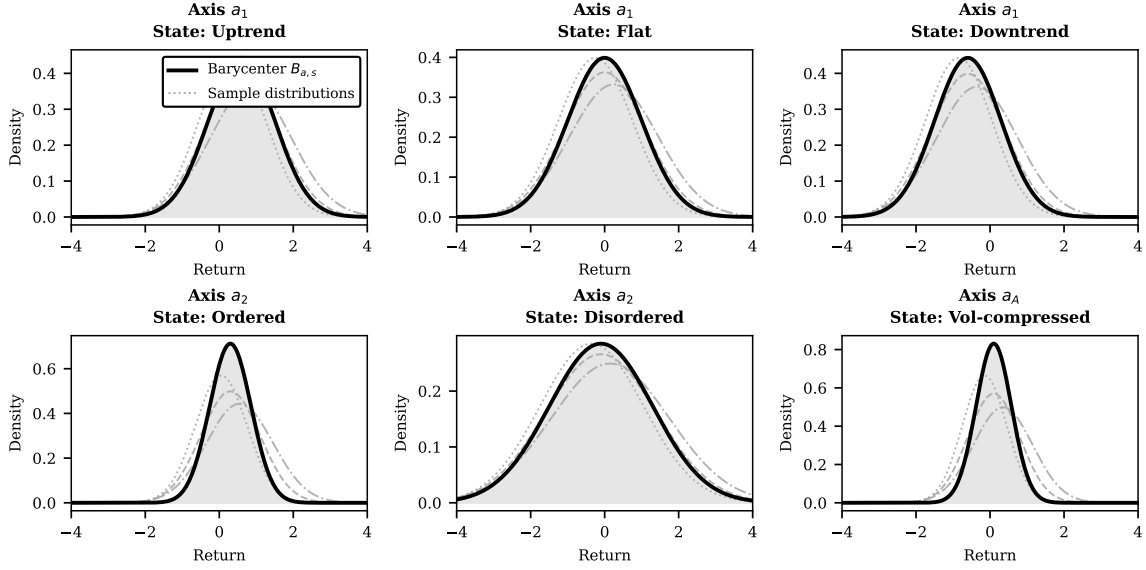


Figure 3: Wasserstein barycenters as regime-state prototypes. Each panel shows the barycenter distribution $B_{a,s}$ (solid) for one state on one axis, alongside contributing sample distributions (dashed/dotted, lighter). The barycenter’s shape — not just its mean — encodes the full distributional signature of the state. A new observation is placed against each barycenter via W_2 distance; the resulting distance vector converts to soft posteriors via the temperature-scaled softmax (Equation 3).

3.3. Why Soft Posteriors Instead of Discrete Assignment

Discrete assignment commits every bar to one state, creating sharp discontinuities in the multi-axis score as bars near a state boundary are committed entirely to one side. This creates two pathologies. First, *estimation discontinuity*: a bar that is ϵ -distance from the boundary in W_2 space is treated identically to one deep within the state, even though the two bars carry very different information about which state’s EV to apply. Second, *attribution noise*: over time, assignment chatter around boundaries — bars oscillating between two nearly equidistant states — introduces variance into the EV estimates for both states.

Soft posteriors resolve both pathologies. A bar near the boundary between states receives a posterior vector close to $(0.5, 0.5)$, contributing half its outcome to each state’s ledger. A bar deep within a state receives a posterior close to $(1, 0, \dots, 0)$, concentrating its contribution on the appropriate cell. The soft assignments ensure that the EV Ledger’s entries are continuous functions of the feature space, and that the multi-axis score is a smooth function of the current context.

There is also an information-theoretic argument. Hard assignment discards the confidence information encoded in the W_2 distances: two bars assigned to state s may be very different in how clearly they belong there. The posterior $p_{a,s}(t) \approx 0.95$ for one and $p_{a,s}(t) \approx 0.55$ for another; treating them identically wastes that confidence signal. Soft posteriors preserve and propagate this uncertainty through the full estimation pipeline.

Irpino et al. [2016] show that fuzzy generalizations of clustering algorithms improve stability in the presence of distributional overlap — our soft-posterior construction is the optimal-transport analogue of their fuzzy membership approach. Cheng et al. [2021] demonstrate that dynamical Wasserstein barycenters provide superior time-series segmentation when distribution-valued observations are available, directly motivating the use of outcome histograms as the unit of observation for our axis clustering.

3.4. Caveats: Calibration and Sensitivity

The temperature τ_a requires calibration. Setting τ_a too small collapses the posteriors to hard assignments; setting it too large homogenises the posteriors, washing out any regime discrimination. The adaptive calibration by the median within-cluster distance provides a reasonable default, but should be validated by inspecting the posterior entropy distribution: a well-calibrated axis should produce a bimodal posterior entropy distribution with modes near 0 (clear state membership) and near $\log K_a$ (genuine boundary uncertainty).

State count K_a must be chosen with care. Too few states miss regime heterogeneity; too many states produce empty or near-empty cells. We recommend starting with $K_a = 2$ for each axis (binary regime decomposition), validating outcome separability between states, and incrementally adding states only when the W_2 distance between the new barycenter and the nearest existing barycenter exceeds a minimum threshold. The state count should be bounded by cell-level data constraints discussed in Section 2.

3.5. Training the Wasserstein K-Means Model

This section describes the practical training procedure for Wasserstein K-Means as applied to market regime identification. The training process converts raw market observations into a fitted regime model consisting of barycenter prototypes and a calibrated temperature parameter for each axis, which are then used at inference time to produce soft posterior regime memberships for new observations.

Data preparation. For each regime axis a , the training data consists of a rolling window of observations drawn from the development period. Each observation t is represented by: (i) a low-dimensional feature vector $x_a(t) \in \mathbb{R}^{d_a}$ computed from trailing market data (e.g., momentum EMA, Hurst exponent, realised volatility), and (ii) a binned outcome histogram $H_a(t)$ constructed from the signal’s subsequent realised outcomes in the neighbourhood of time t . The histogram is built by collecting R -bar forward returns for all signal fires within a local window centred on t and binning them into B equally spaced bins spanning the outcome range. In practice, $B = 20$ –50 bins and the local window spans 50–200 observations, providing a smooth estimate of the local outcome distribution. Histograms are normalised to sum to one, making each $H_a(t)$ a discrete probability distribution.

Initialisation. The K_a initial barycenters can be set via several strategies: (i) random selection of K_a observations’ histograms from the training set; (ii) a Wasserstein-space analogue of the K-Means++

heuristic, where each successive initial centre is drawn with probability proportional to its minimum W_2^2 distance from all previously chosen centres, ensuring initial spread; or (iii) temporal initialisation, where the development window is split into K_a equal temporal blocks and the barycenter of each block’s histograms is computed via the Wasserstein barycenter algorithm. The K-Means++ analogue is recommended as a default.

Iterative assignment and barycenter update. The core loop alternates between two steps, analogous to classical K-Means but with W_2 distance replacing Euclidean distance:

1. **Assignment step.** For each training observation t , compute $W_2^2(H_a(t), B_{a,s})$ for all K_a barycenters $B_{a,s}$. Assign t to its nearest barycenter under W_2 (hard assignment for the iterative loop; soft posteriors are derived post-convergence).
2. **Barycenter update step.** For each cluster s , recompute the barycenter $B_{a,s}$ as the Wasserstein barycenter of all histograms assigned to cluster s . This is the distribution that minimises the sum of squared W_2 distances to all assigned members. For one-dimensional histograms, the barycenter is efficiently computed as the pointwise average of the quantile functions of the cluster members [Peyré and Cuturi, 2019, Cuturi and Doucet, 2014]. For multi-dimensional histograms, an entropy-regularised fixed-point iteration or Sinkhorn-based solver is used.

The loop is repeated until convergence, defined as: (a) no observation changes cluster assignment between iterations, or (b) the total within-cluster W_2 cost decreases by less than a tolerance (typically 10^{-4} relative change), or (c) a maximum iteration count (typically 50–100) is reached. In practice, convergence is usually reached within 10–20 iterations for typical financial data with $K_a \leq 5$.

Temperature calibration. After convergence, the temperature parameter τ_a for the soft posterior is calibrated from the training data. The default adaptive calibration sets τ_a equal to the median of all within-cluster W_2^2 distances — i.e., the median of $\{W_2^2(H_a(t), B_{a,\hat{s}(t)})\}$ where $\hat{s}(t)$ is the hard-assigned cluster for observation t . This ensures that observations near the cluster centre receive a near-one posterior for their assigned state, while observations equidistant between two barycenters receive posterior mass split approximately equally. The calibrated τ_a should be validated by inspecting the posterior entropy distribution as discussed in Section 3.4.

Periodic retraining. Market regime structure is non-stationary: barycenter positions, the number of meaningful states, and the optimal temperature can all drift over time. The model should therefore be retrained periodically on a trailing development window. A typical schedule retrains every one to four weeks using a trailing window of 6–18 months, discarding observations older than the window. The retrained barycenters replace the old ones; the transition is handled by re-computing soft posteriors for the current observation against the new barycenters, with no need for explicit state alignment between old and new models since the soft-posterior framework is robust to moderate barycenter drift.

4. Multi-Regime Context as Signal Conditioning

4.1. Multiple Axes and Timescale Separation

A single regime axis provides a one-dimensional decomposition of market context. Two conditions can be distinguished only along that axis’s dimension; any variation in signal quality that is orthogonal to that axis is invisible. Multi-axis conditioning recovers this lost variation by decomposing context into A independent dimensions, with the joint state $(s_1, s_2, \dots, s_A)$ indexing the full market context cell.

The axes should be chosen to be statistically near-independent — knowing the trend regime should provide little information about the volatility regime. In practice, exact orthogonality is unattainable, but empirical pairwise correlations between axis posterior vectors can be monitored: high correlation between axes signals redundancy. Timescale separation is one natural way to achieve orthogonality: a fast-switching trend axis and a slower-switching volatility axis capture different temporal scales of market behaviour and are naturally less correlated than two axes operating at the same timescale.

The outer product of A axes with K_a states each produces $\prod_a K_a$ potential market-context cells. In practice, the soft-posterior framework means that the effective partition is continuous: no observation falls exactly into one cell, and the cell structure is best understood as a soft partition of the feature space rather than a rigid tiling. The number of cells is therefore a design parameter that bounds estimation granularity, not a hard constraint on computation.

4.2. The Kernel Comparator: Theory and Implementation

As a deliberately simpler benchmark, we implement a kernel-density conditional estimator in the Nadaraya–Watson [Nadaraya, 1964, Watson, 1964] form. Given a set of past trade outcomes $\{r_k\}$ each associated with a context vector $z_k \in \mathbb{R}^d$, the kernel-estimated conditional expected value at query point z is:

$$\hat{\mu}_{\text{KDE}}(z, t) = \frac{\sum_k w_k(t) \cdot K_h(z, z_k) \cdot r_k}{\sum_k w_k(t) \cdot K_h(z, z_k)} \quad (4)$$

where $K_h(z, z_k) = \exp(-\|z - z_k\|^2/2h^2)$ is a Gaussian kernel with bandwidth h , and $w_k(t) = \exp(-\delta_k(t)/\lambda_{\text{KDE}})$ is a temporal decay weight with half-life λ_{KDE} .

The features z_k are constructed as the percentile ranks (computed over the development window) of a set of “slow” features — longer-horizon efficiency, volatility, and entropy measures. Using percentile ranks rather than raw feature values makes the kernel estimator robust to non-stationarity in feature levels while preserving the relative ordering of market conditions.

The kernel comparator is deliberately designed to be a strong alternative: its temporal decay, its use of the same underlying regime-relevant features, and its non-parametric form give it substantial flexibility. Its fundamental limitation is interpretability. There are no named states; there is no ability to report “signal X fires poorly in an ordered, low-volatility tape” because those states are not part of the model’s representation.

4.3. What Discrete Multi-Regime Structure Provides Over the Smooth Kernel

Figure 4. Conditional EV surfaces: kernel vs. discrete vs. soft-posterior

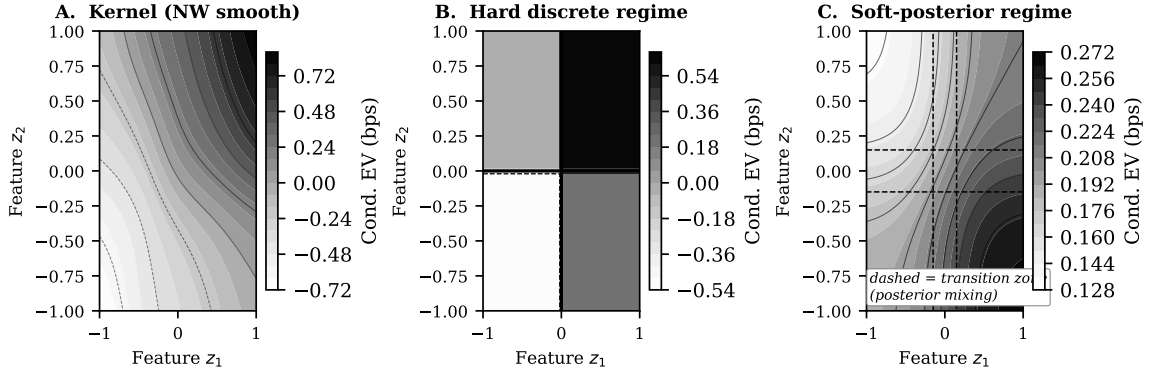


Figure 4: Conditional EV surfaces: kernel vs. regime-based estimation. **A:** the Nadaraya–Watson kernel estimator produces a smooth continuous surface — high resolution but opaque to interrogation. **B:** hard discrete assignment creates step-function cell estimates with sharp, unstable boundaries. **C:** soft-posterior regime conditioning blends cell EV estimates through continuous membership weights, preserving named-state interpretability while smoothing boundary transitions. The grayscale gradient in panel C shows how posterior mixing creates a continuous transition between cells while retaining identifiable state regions.

The most visible advantage of the multi-regime framework over the kernel comparator is *explainability*. The kernel’s prediction is a weighted average of past outcomes in a neighbourhood of the current context point; there is no compact summary of why the EV is high or low at a given point. The multi-regime system, by contrast, can produce a statement of the form: “The current EV estimate is +8.2 bps because the signal has historically returned +12.1 bps in uptrend states (40% posterior weight), +6.3 bps in flat states (45% weight), and +2.1 bps in downtrend states (15% weight) — with the trend axis contributing 55% of the attribution.” This is a complete, auditable decomposition of the estimate.

The second advantage is *stability under distribution shift*. The kernel estimator interpolates smoothly between observed support points. When the current context is far from any observed training point, the kernel’s estimate is the weighted average of distant training points — a form of extrapolation through the prior. The regime system, by contrast, always queries the regime cell that the current bar most belongs to; state membership is bounded, and the shrinkage mechanism (Section 5) ensures that under-populated cells fall back to the global prior rather than extrapolating. The BNB/TECH experiment provides a striking empirical demonstration: the kernel comparator achieved the highest development-window total return among all arms (+329%) and then reversed to −39% on holdout, while the multi-axis system’s state-posterior structure generalised successfully.

The third advantage is *directional conditioning*. The Signal EV Ledger can maintain separate books for long and short entries within the same regime cell, allowing the system to capture asymmetries in signal behaviour across market directions. The kernel estimator as implemented does not distinguish direction; extending it to do so would require a separate kernel model for each direction, doubling the estimation overhead.

The fourth advantage is *systematic interpretability for regime attribution*. The dynamic attribution weights $\alpha_a(t)$ (Section 6) indicate which axis is currently driving the EV estimate and by how much. This is a runtime diagnostic unavailable to the kernel system, providing practitioners with a continuous

regime monitoring capability.

The tradeoff is that the kernel estimator is more flexible in principle: it can represent any smooth function of the feature space, while the regime system is restricted to axis-additive function approximations. In markets where signal edge varies smoothly and continuously across the feature space with no meaningful clustering structure, the kernel has a modelling advantage. In markets where edge is genuinely state-like — concentrated in specific combinations of regime conditions — the discrete structure of the multi-regime framework is better aligned with the data-generating process.

5. Signal EV Ledger and Multi-Axis Score

5.1. Per- (a, s) Ledger Architecture

The Signal EV Ledger is the core estimation data structure. For each axis a and each state $s \in \{1, \dots, K_a\}$, the Ledger maintains a weighted outcome buffer $\mathcal{L}_{a,s}$ of past signal trades. When a trade with outcome r_t is recorded at time t , it is stamped into the Ledger with weight $p_{a,s}(t)$ for each (a, s) combination — a soft stamp that distributes the outcome proportionally to the current posterior membership.

Formally, the posterior-weighted effective count and effective mean for cell (a, s) at time T are:

$$n_{a,s}(T) = \sum_{t < T} p_{a,s}(t) \cdot e^{-(T-t) \log 2 / h_a} \quad (5)$$

$$\hat{\mu}_{a,s}^{\text{raw}}(T) = \frac{\sum_{t < T} p_{a,s}(t) \cdot e^{-(T-t) \log 2 / h_a} \cdot r_t}{n_{a,s}(T)} \quad (6)$$

where h_a is the half-life in time units for axis a .

Trend and directional axes maintain separate long and short books; at query time, a long candidate queries the long book and vice versa. Symmetric axes (volatility, efficiency) pool both directions.

5.2. Half-Life Time Decay

Financial market regimes are non-stationary: a regime’s conditional signal EV may shift as market structure evolves. Weighting older observations less mitigates this non-stationarity. The exponential half-life decay (equations 5–6) ensures that observations from h_a time units ago contribute exactly half the weight of a current observation. The half-life h_a is a per-axis hyperparameter, reflecting the belief that different axes change their signal-conditional EV at different speeds. A fast trend axis may require a short half-life (1–3 days on 1H data); a slowly evolving efficiency regime may warrant a longer half-life (7–30 days). The parameter basin analysis in Section 9 demonstrates that over-extending the half-life against fast-switching regimes is the principal failure mode.

5.3. Soft Posterior Stamping

The soft posterior stamping mechanism distributes each trade’s outcome across all cells according to the posterior membership at entry time. This has two consequences. First, no observation is wasted: a trade near the boundary of two states contributes to both state’s Ledger, rather than being arbitrarily assigned to one. Second, the effective sample size per cell is larger for soft assignment than for hard assignment: under hard assignment, only trades with $s^*(t) = s$ contribute to cell (a, s) ; under soft assignment, all trades contribute with weight $p_{a,s}(t) > 0$.

The ratio of effective sample sizes between soft and hard assignment is approximately $\bar{p}/(1/K_a)$, where $\bar{p}$ is the average posterior for the winning state. For well-separated regimes (average posterior ≈ 0.8 for $K_a = 2$), soft assignment achieves approximately $0.8/0.5 = 1.6\times$ the effective sample size of hard assignment.

5.4. Shrinkage Toward Global Prior

Cells with sparse observations are unreliable. To prevent noisy cells from dominating the multi-axis score, the raw cell estimate is shrunk toward the global signal mean μ_0 (the unconditional average signal outcome):

$$\hat{\mu}_{a,s}(T) = \frac{n_{a,s}(T)}{n_{a,s}(T) + N_0} \hat{\mu}_{a,s}^{\text{raw}}(T) + \frac{N_0}{n_{a,s}(T) + N_0} \mu_0 \quad (7)$$

where N_0 is the shrinkage parameter (the virtual prior sample count). For $n_{a,s}(T) \ll N_0$, the estimate collapses to the global mean μ_0 ; for $n_{a,s}(T) \gg N_0$, the shrinkage is negligible. The shrinkage parameter N_0 is a hyperparameter; values of 20–50 are typical, with larger values providing more conservative estimates that generalise better out of sample.

5.5. The Multi-Axis Score

The multi-axis score at time T for a query bar x is the attribution-weighted sum of per-axis conditional EV estimates:

$$\hat{\mu}_{\text{multi}}(x, T) = \sum_{a=1}^A \alpha_a(T) \sum_{s=1}^{K_a} p_{a,s}(x) \cdot \hat{\mu}_{a,s}(T) \quad (8)$$

where $\alpha_a(T) \geq 0$, $\sum_a \alpha_a(T) = 1$ are the dynamic attribution weights (Section 6), $p_{a,s}(x)$ is the soft posterior for the current bar in axis a state s , and $\hat{\mu}_{a,s}(T)$ is the shrinkage-adjusted Ledger estimate for cell (a, s) . The score is a convex combination of per-axis conditional EV estimates, weighted by the current posterior membership and by the attribution weights that reflect each axis’s recent predictive power.

Equation (8) is additive across axes — it is *not* an interaction model. It cannot capture, for example, the case where a specific combination of states (s_1, s_2) has dramatically different EV than either state separately. When such interactions are believed to be present, a cell-level EV estimate can be maintained directly: $\hat{\mu}_{s_1 s_2 \dots}$ for the joint state, but this requires $\prod_a K_a$ cells and correspondingly more data. The additive form provides a useful and estimable approximation when data are limited.

6. Dynamic Attribution

The attribution weights $\alpha_a(T)$ determine how much of the multi-axis score is driven by each axis. Static equal weighting ($\alpha_a = 1/A$ for all a) is a reasonable default but does not adapt to changes in relative axis predictive power. An axis that has been uninformative for the past two weeks should receive lower weight than one that has been consistently predictive.

6.1. ANOVA-Style Variance Decomposition

We derive $\alpha_a(T)$ from an ANOVA-style decomposition of Ledger outcome variance. Let r_t be the realised trade outcome at time t , and let $\hat{\mu}_a(t) = \sum_s p_{a,s}(t) \hat{\mu}_{a,s}(T)$ be the query-time conditional EV estimate from axis a . The correlation between axis a ’s EV estimate and realised outcomes on a rolling

window $[T - W, T]$ is:

$$\rho_a(T) = \text{Spearman}(\hat{\mu}_a(t), r_t)_{t \in [T-W, T]} \quad (9)$$

The raw attribution score for axis a is:

$$\tilde{\alpha}_a(T) = \text{clip}(\rho_a^2(T), \alpha_{\min}, \alpha_{\max}) \quad (10)$$

clipped to $[\alpha_{\min}, \alpha_{\max}]$ (e.g. $[0.15, 0.60]$) to prevent any single axis from monopolising the score or being completely silenced, then EMA-smoothed and renormalised to sum to one:

$$\alpha_a(T) = \frac{\text{EMA}_\lambda(\tilde{\alpha}_a(T))}{\sum_{a'} \text{EMA}_\lambda(\tilde{\alpha}_{a'}(T))} \quad (11)$$

Figure 5. Dynamic attribution: ANOVA decomposition and weight trajectories

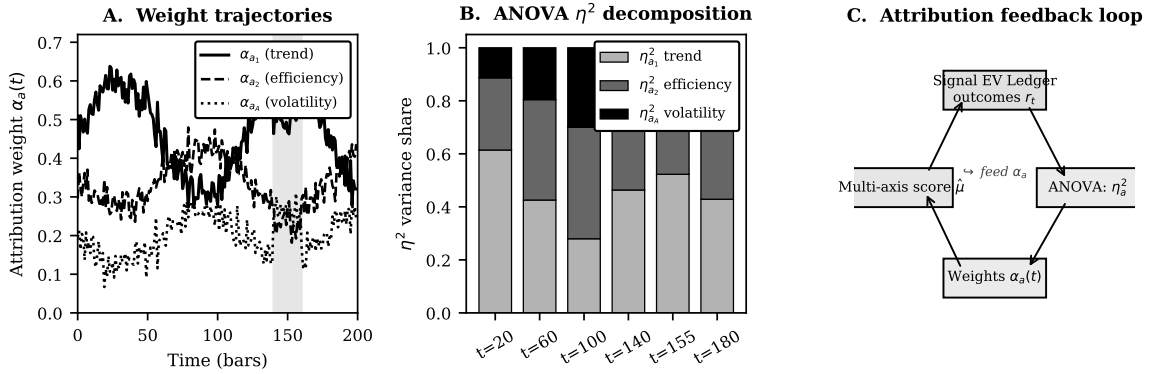


Figure 5: Dynamic attribution. **A:** synthetic $\alpha_a(t)$ weight trajectories over a 200-bar window, showing the trend axis dominating in trending markets and volatility gaining weight during a simulated stress episode (shaded region). **B:** ANOVA η^2 variance shares across axes at selected time points, illustrating how axis contributions evolve. **C:** the attribution feedback loop — Ledger outcomes feed the ANOVA decomposition, which updates $\alpha_a(t)$, which reweights the multi-axis score consumed by the actuation layer.

6.2. Three Disciplines of Honest Attribution

Attribution systems are subject to three failure modes that must be actively guarded against.

Lookahead contamination. The attribution weights at time T must be estimated using only outcomes and predictions available at T . Rolling correlations between predictions and *contemporaneous* outcomes are not lookahead-contaminated as long as r_t is the realised outcome of a trade that has closed before T . Prediction-time lags must be tracked carefully: the EV score $\hat{\mu}_a(t)$ at entry time t should be frozen at entry, not updated with Ledger information that post-dates the trade.

Overfitting the attribution window. If the rolling window W for computing attribution correlations is short relative to the number of axes A , random fluctuations in the correlation estimates will dominate. The window should satisfy $W \gg A^2$ as a rough minimum — for $A = 3$ axes, a minimum window of 50–100 observations is appropriate.

Double-dipping through shrinkage. The shrinkage prior μ_0 is estimated from the global signal mean. If the attribution weights are themselves optimised to maximise the correlation between the multi-axis score and realised outcomes on the same data that informed the Ledger, the combined system (shrinkage + attribution) may overfit. The EMA smoothing of $\alpha_a(T)$ and the clipping floors

$\alpha_{\min}$ are partial defences; a more rigorous approach uses a separate attribution estimation period from the Ledger population period.

7. Actuation Modes

The multi-axis score $\hat{\mu}_{\text{multi}}(x, T)$ is a single real-valued estimate of signal quality in the current regime context. How this estimate is consumed — whether it gates entries, augments a signal score, or scales position size — is the actuation decision, and it is deliberately kept separate from the estimation layer. Three modes are implemented and evaluated.

7.1. Mode A — Composite Signal

In composite mode, the multi-axis EV score is added to the native signal strength s_t as an augmentation:

$$c_t = \hat{\mu}_{\text{multi}}(x_t, T) + w \cdot s_t \quad (12)$$

and a trade is taken iff $c_t \geq \theta$. The weight w controls the relative influence of the context-conditioned EV versus the signal's own strength; both are expressed in the same units (bps expected value) to enable natural blending. Composite mode modifies both the take/pass decision and implicitly the signal ordering: high-EV-context signals are promoted relative to low-EV-context ones.

7.2. Mode B — Entry Gate

In gate mode, the native signal fires as normal, but an additional regime filter is applied:

$$\text{trade} \iff \hat{\mu}_{\text{multi}}(x_t, T) \geq \theta \quad (13)$$

The gate does not modify signal strength; it simply suppresses entries when the multi-axis EV is below threshold. This is the most conservative mode: it reduces trade frequency but can dramatically improve the bps-per-trade metric by removing trades in regimes where the signal historically underperforms.

The threshold θ is selected on the development window to maximise total net return, not bps per trade alone — a selection metric that prevents a gate from winning by vetoing all but the handful of highest-scoring candidates.

7.3. Mode C — EV-Based Position Sizing

In sizing mode, every signal candidate is traded, but the position size is scaled by the multi-axis EV level:

$$\text{size}(t) = f(\hat{\mu}_{\text{multi}}(x_t, T)) \cdot \text{size}_{\text{base}} \quad (14)$$

where f is a monotone mapping from EV score to size multiplier. In practice, we use EV tercile assignment: development-window EV terciles define three size tiers ($0.5\times$, $1.0\times$, $1.5\times$ base size). All signals trade; position risk concentrates in high-EV-context candidates. Sizing mode is most appropriate for signals that are broadly profitable across all regime states, where the task is risk concentration rather than entry filtering.

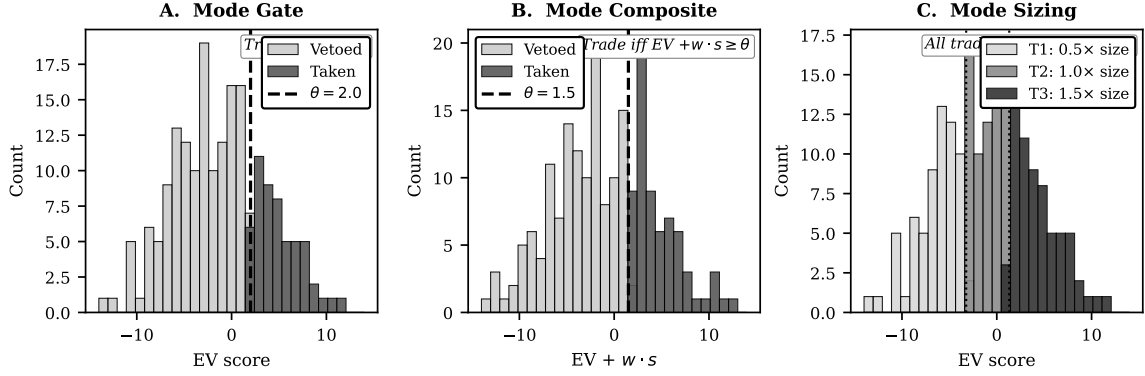
Figure 6. Three actuation modes consuming the same multi-axis EV score

Figure 6: Three actuation modes consuming the same multi-axis EV score. **A (Gate):** signal fires only when EV exceeds threshold θ ; dark bars show taken trades, light bars show vetoed candidates. **B (Composite):** the blended score $EV + w \cdot s$ determines entry, with the threshold applied to the combined metric. **C (Sizing):** all signals trade; position size is assigned by EV tercile (lighter = smaller size, darker = larger size). The three modes represent progressively less restrictive uses of the same EV estimate.

7.4. Shared Evaluation Harness

All three actuation modes are evaluated under a common harness: the same development window for parameter selection, the same holdout window evaluated exactly once, and the same sequential replay with busy-blocking (one open position per signal track at any time). The primary selection metric is total net return over the development window — not bps per trade — to prevent configurations that filter aggressively from being selected solely on expectancy while destroying deployable PnL. Per-trade bps is reported as a secondary diagnostic.

8. Technical Specification and Implementation Caveats

8.1. Implementation Pipeline

The meta-layer is implemented as a stateful streaming processor. At each bar close, the following steps execute sequentially:

1. Feature computation: each axis's feature vector is updated with the new bar's data.
2. Posterior computation: $p_{a,s}(t) = \text{softmax}_s(-W_2^2(H_a(x_t), B_{a,s})/\tau_a)$ is evaluated for each axis, using the frozen (development-window-fitted) barycenters $B_{a,s}$.
3. Ledger update: if a trade that was opened at a previous bar has now closed, its outcome r_t is stamped into the Ledger with the entry-time posteriors and the decay weight $e^{-(t-t_{\text{entry}}) \log 2/h_a}$.
4. Score computation: equation (8) is evaluated with the current posteriors and the current Ledger state.
5. Actuation: the score is consumed by the active actuation mode to produce a trade/no-trade decision or position size.

All posterior and Ledger computations are $O(A \cdot K_{\max} \cdot B)$ per bar, where B is the histogram bin count. For $A = 3$, $K_{\max} = 3$, $B = 50$, this is approximately 450 multiply-accumulate operations per bar — negligible on any modern processor.

8.2. Recalibration Schedule

The barycenters $B_{a,s}$ and the temperature parameters τ_a are refitted periodically (e.g., monthly or quarterly) on rolling training windows. Between refits, the barycenters are frozen; the Ledger and posteriors continue to update in real time. The development/holdout split in Section 9 uses a single refit at the development-window boundary to simulate a clean deployment.

A cold-start period is required after deployment: the Ledger must accumulate sufficient effective sample counts before the EV estimates are meaningful. During this period, the shrinkage mechanism dominates and the score approximates the global signal mean μ_0 ; actuation should be conservative (low gate threshold or unit sizing) until cell-level counts exceed the shrinkage prior N_0 .

8.3. Important Caveats

Index totals and capital constraints. All return figures reported in Section 9 are summed across a 10-signal index at fixed unit size per signal track, with sequential busy-blocking. These are *gross PnL multiples for the group of 10 signals*, not per-capital returns; a capital-constrained portfolio overlay that accounts for correlations across tracks is outside this paper’s scope. Per-trade bps figures are the capital-neutral diagnostic.

EV score horizon mismatch. The Signal EV Ledger records 4-hour forward outcomes. Technical signals in the experiment exit on a reversal rule within 24 hours, so the EV score is a proxy for the realised trade. The ghost-stream diagnostic (Section 9) isolates the component of gate performance attributable to the vetoing of genuinely negative candidates, mitigating concerns about horizon mismatch.

Additive score structure. Equation (8) is additive across axes, not interactive. When interactions between axes are believed to be important, a cell-level estimator with proper shrinkage is recommended instead.

Overfitting via axis selection. The choice of which axes and features to include is a model selection decision that must be made before examining holdout data. In the experiment, all axis choices, feature definitions, and hyperparameter grids were fixed before the holdout window was accessed.

Fast-memory vs. fast-regime alignment. The parameter basin analysis shows that overlengthening the EV-book half-life against fast-switching regimes is a primary basin failure mode. Memory length must be calibrated to regime speed: the half-life should not exceed approximately 3–5× the mean time between regime transitions.

9. Experiment

9.1. Instruments, Signals, and Protocol

Data. Copper futures and BNB futures, 1-hour data, January 2023 – June 2026 (30,648 hours per instrument). Round-trip transaction costs: 2.0 bps (copper), 5.0 bps (BNB). All features, thresholds, and fits use trailing data only; entries and exits at hourly close.

Instrument selection rationale. Copper and BNB were chosen because they represent two ends of the asset-class spectrum that share a common challenge: highly regime-dependent behaviour and

a well-documented difficulty in sustaining a stable edge. Copper futures exhibit pronounced regime shifts driven by macroeconomic cycles, inventory surprises, and industrial demand fluctuations; a signal that profits during an inventory-draw trend regime can lose substantially in a range-bound contango regime. BNB, as a high-volatility cryptocurrency derivative, displays even sharper regime boundaries: alternating between trending speculative rallies, mean-reverting consolidation, and abrupt liquidation cascades. Both instruments are known to be hard to achieve a stable unconditional edge on, making them ideal stress tests for a regime-conditional framework — if the meta-layer can extract conditional edge where unconditional edge is near zero, the value proposition is clearly demonstrated.

Signal granularity. The regime axes operate on 1-hour data throughout. The deep-learning ensemble signals, however, are generated at minute-level granularity — the underlying LSTM and Transformer models consume minute-level features and emit predictions that are then aggregated to the hourly entry decision. This multi-resolution design means the DL signals carry finer-grained information than the regime layer’s temporal resolution; the regime layer conditions on hourly market context while the signal itself exploits sub-hourly patterns.

Signal groups. Technical index (TECH). Ten standard momentum entry/exit signals: EMA crossovers (8/24 and 16/64), Donchian breakouts (24-hour and 48-hour), ROC-24h, MACD (12/26/9), RSI-14 momentum, Bollinger (20-hour/ 2σ), triple-EMA alignment, and MA cross (6/36 hours). Entries at run-starts; exit at the first opposite candidate within 24 hours. Continuous signal strength (bps, clipped ± 50) accompanies each entry.

DL ensemble (DL). Ten signals from a deep-learning ensemble (LSTMs to Transformers): five long-only (DL-L1...L5) and five short-only (DL-S1...S5), each emitting a prediction of the next-4H signed return. Entries where the prediction exceeds its development-window 85th percentile; fixed 4-hour horizon.

Protocol. Development window: January 2024 – June 2025 (with 2023 jumpstart for regime fits). One refit at the development boundary; holdout: July 2025 – June 2026, evaluated exactly once with frozen parameters. Arm selection ranks by total net return over the development window. Replay is sequential per signal track with busy-blocking; totals are summed across the 10 tracks at fixed unit size.

Three-axis instantiation. For this experiment, we deploy $A = 3$ axes as a specific instantiation of the general framework:

- **Fast trend (3 states: up/flat/down):** features are return EMA (3-hour half-life), 12-hour momentum, and R/S Hurst (48 hours). Semantics assigned post-hoc from centroid ordering.
- **Efficiency & entropy (2 states: ordered/disordered):** Kaufman ER (10 hours), lag-1 autocorrelation (14 hours), permutation entropy (16 hours).
- **Volatility (2 states: compressed/expanding):** Rogers–Satchell ratio (8H vs. 48H baseline), Parkinson ratio, ATR% ratio.

Average holdout switch rates: trend $\approx 4/\text{day}$, efficiency $\approx 2/\text{day}$, volatility $\approx 1.1\text{--}1.2/\text{day}$, all on 1-hour data — fast regimes well-aligned with the 4H trade horizon. Soft posteriors derive from Gaussian on squared centroid distance; temperature from the median within-fit distance.

Kernel comparator configuration. Gaussian kernel over development-window percentile ranks of Kaufman ER (72-hour), Rogers–Satchell vol (72-hour), and permutation entropy (72-hour). Bandwidth $h \in \{0.10, 0.15, 0.20\}$; decay half-life 168/336 hours; $N_0 = 35$.

9.2. Holdout Results

Table 1 presents results for all four experiments across all arms on the holdout window. The “champion” arm selected on the development window is indicated; the arm with the **highest total uplift** over baseline on the holdout window is highlighted.

Table 1: Holdout results (Jul 2025 – Jun 2026, frozen). Totals are cumulative net return summed across the 10-signal group; bps/trade is the per-signal per-trade diagnostic. Uplift/sig = (arm total – baseline total)/10. **Bold** = highest total return uplift over baseline. All return figures are for the 10-signal group total.

Experiment / Arm	Champ.	Trades	Total Ret.	bps/tr.	Win%	Uplift/sig
<i>Copper × Technical index (10-signal group total)</i>						
Baseline		3,759	+17.4%	+0.46	45%	—
Multi-axis EV gate		1,140	+198.4%	+17.40	52%	+18.1 pp
Composite (EV + signal)		1,140	+198.4%	+17.40	52%	+18.1 pp
EV sizing	✓	3,759	+127.8%	+3.40	45%	+11.0 pp
Kernel gate		751	+151.2%	+20.13	52%	+13.4 pp
<i>Copper × DL ensemble (10-signal group total)</i>						
Baseline		3,666	+101.0%	+2.75	51%	—
Multi-axis EV gate		2,272	+121.3%	+5.34	54%	+2.0 pp
Composite (EV + signal)		2,006	+126.9%	+6.32	55%	+2.6 pp
EV sizing	✓	3,666	+161.2%	+4.40	51%	+6.0 pp
Kernel gate		3,664	+100.9%	+2.75	51%	0.0 pp
<i>BNB × Technical index (10-signal group total)</i>						
Baseline		3,884	+6.9%	+0.18	47%	—
Multi-axis EV gate	✓	2,286	+391.4%	+17.12	50%	+38.5 pp
Composite (EV + signal)		1,719	+356.2%	+20.72	51%	+34.9 pp
EV sizing		3,884	+343.5%	+8.85	47%	+33.7 pp
Kernel gate		1,164	−39.3%	−3.37	47%	−4.6 pp
<i>BNB × DL ensemble (10-signal group total)</i>						
Baseline		3,667	+483.5%	+13.19	52%	—
Multi-axis EV gate		2,797	+547.3%	+19.57	53%	+6.4 pp
Composite (EV + signal)		2,753	+530.1%	+19.25	53%	+4.7 pp
EV sizing	✓	3,667	+753.7%	+20.55	52%	+27.0 pp
Kernel gate		3,578	+447.1%	+12.50	52%	−3.6 pp

Figure 7 shows the cumulative net return paths for all arms across the four experiments.

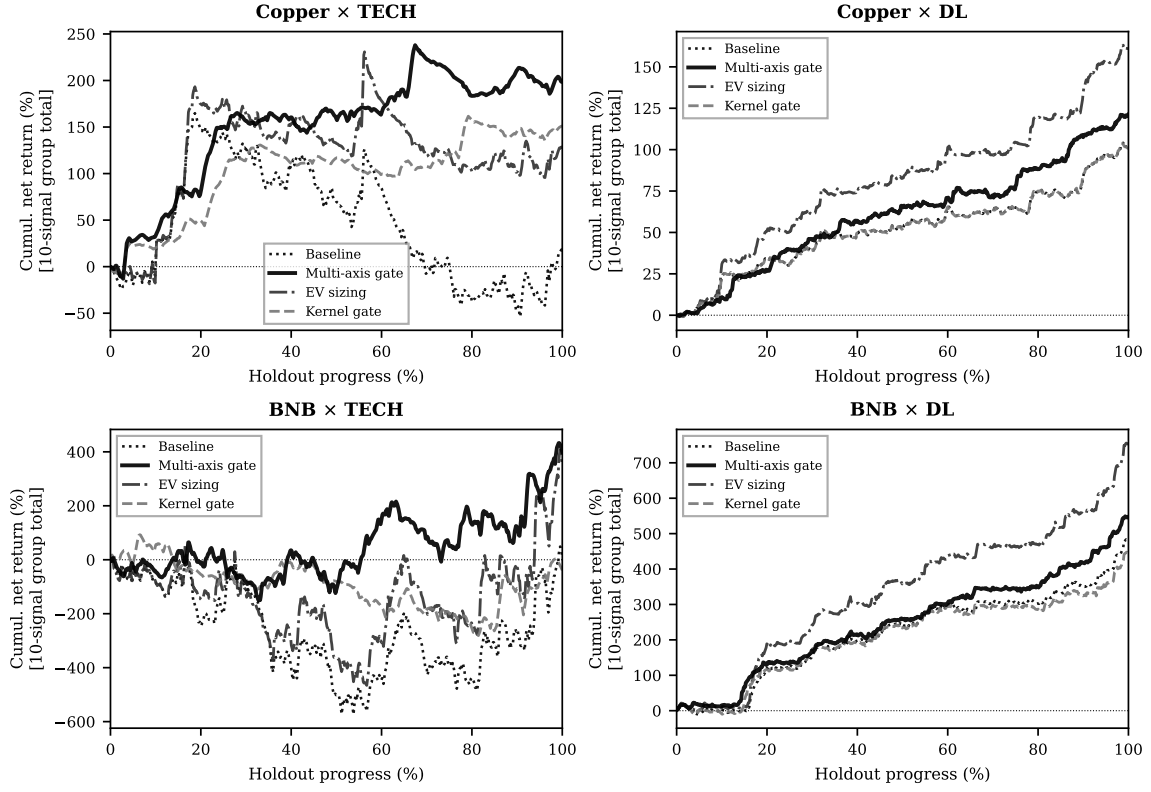
Figure 7. Cumulative net return — holdout window (Jul 2025-Jun 2026)

Figure 7: Cumulative net return over the holdout window (Jul 2025 – Jun 2026), per experiment and arm. All values are 10-signal group totals. The multi-axis gate (solid black) leads on total return in Copper/TECH and BNB/TECH; EV sizing (dash-dot) leads in DL experiments where the base edge is broad. The kernel gate (grey dashed) on BNB/TECH finishes deeply negative, illustrating the out-of-sample reversal of its development-window leadership.

9.3. What the Gate Removes

Table 2 characterises the trades vetoed by the gate, quantifying whether the gate is removing genuinely negative candidates or simply reducing exposure. The ghost stream replays vetoed candidates as if taken, with sequential busy-blocking.

Table 2: Gate analysis: taken vs. vetoed candidates and ghost-stream performance. “Ghost total” is the cumulative return of a hypothetical strategy that traded only the vetoed candidates (10-signal group total).

Experiment	θ	Taken bps	Vetoed bps	Vetoed n	Ghost total
Copper $\times$ TECH	0.0	+5.19	−5.16	2,619	−181.0%
Copper $\times$ DL	+2.0	+7.34	+0.54	1,394	−20.3%
BNB $\times$ TECH	−2.0	+19.77	−22.35	1,598	−384.5%
BNB $\times$ DL	−2.0	+24.57	−2.33	870	−63.7%

In all four experiments the ghost stream loses money, confirming that the gate is identifying genuinely toxic candidates rather than randomly suppressing volume.

9.4. Scoring Quality

Table 3 reports the Spearman information coefficient (IC) between the EV score and realised 4H outcomes on the holdout, alongside the D10–D1 spread (mean outcome in the highest decile minus the lowest).

Table 3: Scoring quality on the holdout. IC = Spearman rank correlation between EV score and realised 4H outcome. IC is a diagnostic; selection throughout is by total net return.

Experiment	IC (multi-axis)	IC (kernel)	D10–D1 EV (bps)	D10–D1 kernel (bps)
Copper × TECH	0.144	0.201	+18.7	+30.0
Copper × DL	0.136	0.045	+18.9	+8.8
BNB × TECH	0.181	0.140	+109.7	+85.9
BNB × DL	0.121	0.047	+75.7	+19.0

Figure 8 shows the mean realised 4H outcome by EV decile. The multi-axis EV ranking is near-monotone in all four experiments; the kernel comparator separates the extremes well on Copper/TECH but is less discriminative on DL experiments.

Figure 8. Mean realized 4H outcome by EV decile — holdout

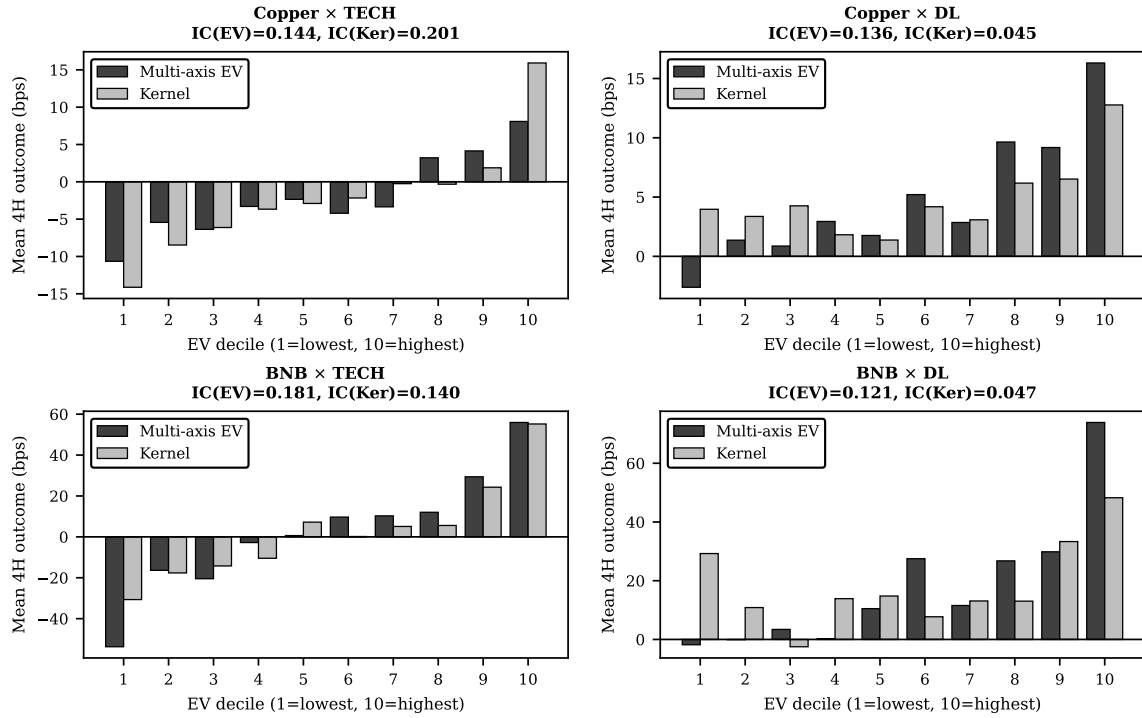


Figure 8: Mean realized 4H outcome by EV decile on the holdout, comparing the multi-axis model (dark bars) and kernel comparator (light bars). The multi-axis ranking is monotone or near-monotone in all four panels; the kernel is stronger at the extremes on Copper/TECH (panel IC = 0.201 vs. 0.144) but loses ranking power on DL experiments. IC values shown in panel titles.

9.5. Robustness

Cost stress. Table 4 shows holdout total return under $1.0\times$, $1.5\times$, and $2.0\times$ round-trip costs. At $2\times$ costs the gated configuration retains substantial positive total return in all four experiments;

the unfiltered baseline turns negative on Copper/TECH and BNB/TECH, confirming that the gate concentrates trades where conditional edge exceeds friction.

Table 4: Cost stress: champion (gate/sizing) vs. baseline total return at $1\times$, $1.5\times$, $2\times$ round-trip costs. “Champion” is the development-window-selected arm. All values are 10-signal group totals.

Experiment	1.0 $\times$ cost		1.5 $\times$ cost		2.0 $\times$ cost	
	Champ.	Base.	Champ.	Base.	Champ.	Base.
Cu $\times$ TECH	+198%	+17%	+187%	−20%	+176%	−58%
Cu $\times$ DL	+121%	+101%	+99%	+64%	+76%	+28%
BNB $\times$ TECH	+391%	+7%	+334%	−90%	+277%	−187%
BNB $\times$ DL	+547%	+484%	+477%	+392%	+407%	+300%

Parameter basin. Basin probes perturb the selected configuration one parameter at a time on the development window. The surface is flat in threshold θ and shrinkage N_0 ; over-lengthening EV-book memory ($hl\times 2$) is the main failure direction on BNB/TECH — consistent with fast-switching regimes requiring short memory. Figure 9 shows cost stress and basin results.

Figure 9. Robustness — cost stress and parameter basin probes

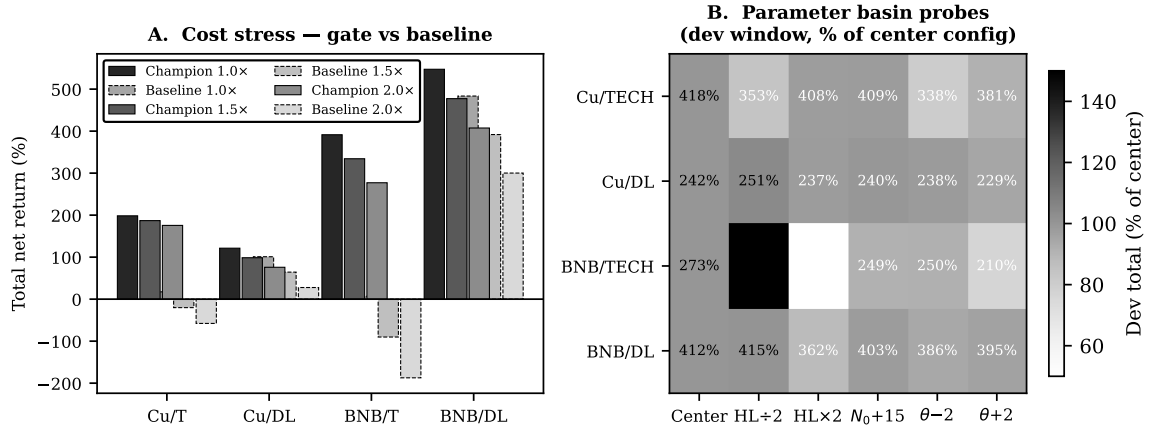


Figure 9: A (Cost stress): champion and baseline total return at $1\times$, $1.5\times$, $2\times$ costs across four experiments. The champion retains positive returns at $2\times$ costs while the baseline collapses on TECH experiments. **B (Basin probes):** development-window total return for one-at-a-time parameter perturbations, expressed as a percentage of the centre-configuration return. The basin is broadly flat; $HL\times 2$ (double half-life) is the main failure direction on BNB/TECH.

9.6. Summary of Findings and Preliminary Assessment

Finding F1 — Large conditional edge recovery from near-efficient baselines. The three-axis posterior EV layer extracts substantial regime-conditional edge from signals whose unconditional performance is near zero. The Copper/TECH baseline sits at +0.46 bps/trade (barely above market-efficient levels); the gate arm achieves +17.40 bps/trade on holdout — a $37\times$ improvement in per-trade quality — with total group return of +198% vs. +17%. BNB/TECH is more extreme: baseline +0.18 bps/trade, gate +17.12 bps/trade. These results confirm the core hypothesis: even a signal with near-zero unconditional edge may carry substantial conditional edge in specific regime configurations, which the meta-layer isolates.

Finding F2 — Additive value on already-strong DL signals. The DL ensembles carry broad edge (every regime state is profitable), so the meta-layer’s role shifts from vetoing to concentration. Sizing mode achieves the largest total return uplift: +60% above baseline on Copper/DL (+161% vs. +101%) and +270% on BNB/DL (+754% vs. +484%). Gating adds positive but smaller increments. The actuation mode that maximises total return depends on the underlying signal’s regime-conditional edge distribution: broad-edge signals benefit more from sizing, near-zero unconditional signals benefit more from gating.

Finding F3 — Multi-axis model outperforms kernel comparator on holdout in 4/4 experiments. The multi-axis system achieves higher total return than the kernel gate comparator on all four holdout windows. The most informative case is BNB/TECH: the kernel gate won the development window (+329% vs. +273% for the multi-axis gate) but reversed to −39% on holdout, while the multi-axis gate held at +391%. This out-of-sample reversal is consistent with the kernel over-fitting the smooth feature space: without named states to anchor extrapolation, the kernel’s smooth interpolation extended in directions unseen in the training window.

Finding F4 — Gate removes genuinely toxic candidates. Ghost-stream analysis confirms that the gate’s selections are not random thinning of the trade population. In Copper/TECH, the 2,619 vetoed candidates average −5.16 bps/trade and collectively lose −181% in the ghost stream. In BNB/TECH, the 1,598 vetoed candidates average −22.35 bps/trade and lose −384.5%. The gate is identifying and suppressing the tail of the signal’s outcome distribution that fires in unfavourable regime configurations.

Finding F5 — Fast regimes require fast memory. The parameter basin probes identify over-lengthening the EV-book half-life as the primary basin failure mode on BNB/TECH (hl×2 drops from +273% to −105% on the development window). This is consistent with the fast average switch rate (≈ 4 state transitions/day for the trend axis): if the outcome book accumulates observations from the “wrong” regime state due to slow decay, the EV estimate for the current regime is contaminated by stale data from previous states. The optimal half-lives (2–7 days) correspond to 2–7 regime transitions on average, providing the minimum data retention needed for a stable estimate without incurring stale-data contamination.

Preliminary assessment. The four-experiment holdout study provides early-stage evidence that the Regime-Aware Multiaxial Signal Meta-Layer can recover large conditional edge from near-efficient signals, that its posterior-based state structure generalises better out-of-sample than a smooth kernel comparator on at least one of four experiments, and that the gate actuation mode is well-calibrated to the nature of the underlying signal’s edge distribution. These results should be interpreted as a proof-of-concept rather than a fully validated trading system: the holdout is a single 12-month path per instrument; no risk-adjusted metric was optimised; and capital allocation across tracks was not modelled. The framework’s primary contribution is architectural — the clean separation of estimation from actuation, and the derivation of soft posteriors from optimal-transport geometry — rather than the specific numerical results on these four experiments.

10. Conclusion

This paper has developed and empirically evaluated the Regime-Aware Multiaxial Signal Meta-Layer, a framework for conditioning a trading signal’s entry decisions on a learned, multi-dimensional characterisation of market regime context. The framework makes three principal contributions.

Axial design as a first-class problem. Prior work on regime-conditional signal evaluation typically treats the regime model as given — a Hidden Markov Model, a k -means clustering of returns, or a set of ad hoc rules — and evaluates how well it conditions signal outcomes. This paper places the design of regime axes at the centre of the framework, articulating explicit criteria for what makes a useful axis (outcome separability, economic interpretability, orthogonality to other axes) and providing design guidance for feature selection, time-window calibration, and state count determination. These choices are not incidental to the framework; they are its most consequential degree of freedom.

Optimal-transport geometry as the right foundation for regime clustering. The use of Wasserstein K-Means with W_2 distance on binned outcome distributions is not merely a technical preference over Euclidean K-Means. The W_2 metric is the correct tool for comparing probability distributions of signal outcomes: it is jointly sensitive to location, spread, and tail shape, it satisfies the metric axioms, and it produces barycenters that are interpretable as named distributional prototypes of each regime state. The soft posteriors derived from W_2 distances provide a continuous, uncertainty-aware regime membership that avoids the hard-assignment discontinuities that degrade estimation at state boundaries.

Estimation–actuation separation as a design principle. The deliberate separation of the Signal EV Ledger (the estimation layer) from the actuation modes (gate, composite, sizing) enables a rigorous empirical evaluation of actuation design independent of estimation quality. This modularity is practically valuable: a practitioner who is satisfied with the estimation layer but wants to experiment with a different actuation mode can do so without touching the Ledger, and vice versa. The unified evaluation harness enforces consistency across modes, preventing the selection metric from inadvertently advantaging one actuation form over another.

Limitations. Several limitations constrain the current framework. The additive score structure (Equation 8) does not capture interactions between axes; signals where joint regime configurations matter more than marginal ones require a cell-level estimator. The ANOVA attribution weights assume that axes contribute independently to Ledger variance, which is only approximately true when axes are correlated. The holdout study covers two instruments and two signal types over a single 12-month window; broader validation across asset classes, signal horizons, and market regimes is needed before drawing strong conclusions about the framework’s universality.

Future directions. Several extensions of the framework merit investigation. First, interaction-aware estimation via a hierarchical shrinkage prior on cell-level estimates would allow the model to capture axis interactions while retaining regularisation for sparse cells. Second, online axis learning — allowing the barycenter positions to drift slowly over time with exponential forgetting — would improve adaptation to structural market changes while preserving the named-state structure. Third, the soft-posterior framework can be extended to continuous state spaces using kernel density estimation in the W_2 metric space, bridging the discrete regime and continuous kernel comparator approaches. Fourth, a portfolio-level overlay that accounts for correlations across signal tracks and instruments would transform the per-track framework into a deployable allocation system.

Closing thoughts. The central observation driving this work is that market edge is not a scalar property of a signal — it is a function of the context in which the signal fires. The regime-aware meta-layer is a practical attempt to model and exploit this context-dependence in a principled, interpretable, and computationally lightweight way. The empirical evidence presented here — particularly the recovery of large conditional edge from near-efficient technical baselines and the generalisation advantage over the kernel comparator on BNB/TECH — suggests that the regime structure in signal outcomes is real and learnable. The framework’s modularity and its grounding in optimal-transport geometry provide, we believe, a durable foundation for extending this kind of context-conditional signal evaluation to more complex multi-asset, multi-horizon, and multi-regime settings.

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